



## SHANTO MARIAM UNIVERSITY OF CREATIVE TECHNOLOGY

**Course Code: CSE 4114**

Course Title: Mobile Application and Development

**SOFTWARE REQUIREMENTS SPECIFICATION (SRS)**

# CSE Club Hub

*A Centralized Mobile Platform for CSE Department Clubs*

Platform: Flutter | Backend: Firebase

|                                                                                                                     |                                                                                           |
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## 1. Introduction

### 1.1 Purpose

This Software Requirements Specification (SRS) document defines the complete functional and non-functional requirements for CSE Club Hub — a cross-platform mobile application developed using Flutter and Firebase for the Computer Science and Engineering department of Shanto Mariam University of Creative Technology.

The purpose of CSE Club Hub is to solve the fragmented communication problem that exists across the department's six active student clubs. Currently, club notices, event announcements, and community updates are scattered across WhatsApp groups, Facebook pages, and physical notice boards — leading to missed information, poor student engagement, and no structured way to distinguish club executives from regular students.

This application provides a unified, role-based digital platform where executives can publish notices and manage events, and students can follow clubs and receive real-time updates — all from a single mobile app.

## 1.2 Scope

CSE Club Hub is a standalone mobile application covering all six clubs of the CSE department. The following defines what the system will and will not do:

### *In Scope:*

- User registration, login, and role-based profile management.
- Six dedicated club spaces with individual feeds, event boards, and member lists.
- Post creation (text and image) by authorized club executives only.
- Real-time push notifications for new posts, events, and reminders.
- Event creation, RSVP management, and automated 24-hour reminders.
- Search for posts, events, clubs, and users across the department.
- Super Admin panel for managing executives and moderating content.

## 1.3 Objectives

1. Build a Flutter-based cross-platform mobile application for Android and iOS.
2. Integrate all six CSE department clubs into one unified, role-based platform.
3. Implement a three-tier role system: Super Admin, Club Executive, and Regular Student.
4. Enable real-time push notifications for club announcements and upcoming events.
5. Provide club executives with tools to post notices (text and image) and manage events.
6. Allow students to follow specific clubs and receive a personalized activity feed.
7. Deploy the application to the Google Play Store for genuine university-wide usage.

## 1.4 Target Users

| User Type        | Description                                                                                                    | Estimated Count             |
|------------------|----------------------------------------------------------------------------------------------------------------|-----------------------------|
| Regular Students | All CSE & CSIT department students who use the app to follow clubs, view notices, and attend events.           | 300–500 students            |
| Club Executives  | President, Vice President, and 3–4 executive members of each of the six clubs. They create and manage content. | ~36 executives (6 per club) |
| Super Admin      | A designated department representative who manages executive roles and moderates the platform.                 | 1–2 admins                  |

## 1.5 Definitions, Acronyms, and Abbreviations

| Term      | Definition                                                                                                  |
|-----------|-------------------------------------------------------------------------------------------------------------|
| SRS       | Software Requirements Specification — this document                                                         |
| CSE       | Computer Science and Engineering                                                                            |
| CSIT      | Computer Science and Information Technology                                                                 |
| Flutter   | Google's open-source UI framework for building cross-platform mobile apps using Dart                        |
| Firebase  | Google's Backend-as-a-Service (BaaS) platform used for authentication, database, storage, and notifications |
| Firestore | Firebase's real-time NoSQL cloud database used to store and sync app data                                   |
| FCM       | Firebase Cloud Messaging — the service used to send push notifications to mobile devices                    |
| RSVP      | Répondez s'il vous plaît — used here to indicate a user's response to an event (Going / Interested)         |
| FR        | Functional Requirement                                                                                      |
| NFR       | Non-Functional Requirement                                                                                  |
| UI        | User Interface                                                                                              |
| API       | Application Programming Interface                                                                           |
| SSO       | Single Sign-On                                                                                              |
| Admin     | Super Administrator of the platform                                                                         |
| Executive | An authorized club member with content creation privileges                                                  |

## 2. Overall Description

### 2.1 Product Perspective

CSE Club Hub is a standalone mobile application that does not replace or integrate with any existing university system. It operates independently using Firebase as its cloud backend, which provides

real-time data synchronization, authentication, file storage, and push notification delivery.

The app interacts with the following external systems and services:

- Firebase Authentication — for secure user login and session management.
- Firebase Firestore — for storing all posts, events, club data, and user profiles in real time.
- Firebase Storage — for hosting uploaded images (post images, avatars, club covers).
- Firebase Cloud Messaging (FCM) — for delivering push notifications to Android and iOS devices.

The application is designed to be deployed on the Google Play Store (Android) as the primary platform, with iOS support available through the same Flutter codebase.

2.2 Product Functions

The following are the primary functions of CSE Club Hub:

| Module             | Key Functions                                                                      |
|--------------------|------------------------------------------------------------------------------------|
| Authentication     | User registration, email/password login, role-based session, profile setup         |
| Club Feed          | View personalized feed, post notices with text/image, pin posts, react and comment |
| Club Management    | Browse all six clubs, follow/unfollow, view club profiles and executive lists      |
| Events             | Create events, view event calendar, RSVP (Going/Interested), receive reminders     |
| Notifications      | Real-time push alerts for new posts, events, comments, and reminders               |
| Search & Discovery | Search posts, events, clubs, and users across the entire platform                  |
| Admin Panel        | Assign/remove executives, moderate content, manage user roles                      |

2.3 User Classes and Characteristics

| User Class      | Characteristics                                                      | Technical Level | Primary Actions                                  |
|-----------------|----------------------------------------------------------------------|-----------------|--------------------------------------------------|
| Super Admin     | 1–2 designated dept. reps. Manages the entire platform.              | High            | Assign executives, remove content, manage users  |
| Club Executive  | ~6 per club (President, VP + members). Responsible for club content. | Medium          | Create posts, manage events, update club profile |
| Regular Student | 300–500 students. Consumers of club content.                         | Low–Medium      | Follow clubs, view feed, react, comment, RSVP    |

2.4 Operating Environment

| Component             | Specification                                              |
|-----------------------|------------------------------------------------------------|
| Primary Platform      | Android 10 (API Level 29) and above                        |
| Secondary Platform    | iOS 13 and above (same Flutter codebase)                   |
| Development Framework | Flutter 3.x (Dart language)                                |
| Backend Services      | Firebase (Firestore, Auth, Storage, FCM)                   |
| State Management      | Riverpod                                                   |
| IDE                   | Android Studio / VS Code                                   |
| Target Devices        | Smartphones (Android and iOS)                              |
| Internet              | Required for all real-time features; no offline mode in v1 |
| Minimum RAM           | 2 GB RAM on device                                         |

2.5 Design and Implementation Constraints

- The application must be developed using the Flutter framework (Dart language).
- Firebase must be used as the sole backend service (no custom server in v1).
- The app must run on Android 10 or above as the minimum supported version.
- Development is carried out by a single student developer within an 8-week timeline.

- All uploaded media must be stored via Firebase Storage (no third-party CDN in v1).
- Role assignment (executive promotion) can only be performed by the Super Admin.
- The application must not exceed 50 MB in APK size for Play Store compatibility.

## 2.6 Assumptions and Dependencies

- All users must have an active internet connection to use the application.
- Firebase servers must be operational for authentication, data sync, and notifications.
- Users must provide a valid university email address during registration.
- Club executives are pre-identified by the department and assigned by the Super Admin before the app goes live.
- The six club names and focus areas remain fixed for the scope of this version.
- FCM delivery depends on the user's device notification permissions being enabled.

## 3. Specific Requirements

### 3.1 Functional Requirements

#### *3.1.1 Authentication & User Management*

| ID    | Requirement Description                                                                                               | Priority |
|-------|-----------------------------------------------------------------------------------------------------------------------|----------|
| FR-01 | The system shall allow new users to register using a university email address and password.                           | High     |
| FR-02 | The system shall validate that the email follows the university domain format during registration.                    | High     |
| FR-03 | The system shall allow registered users to log in using their email and password.                                     | High     |
| FR-04 | The system shall allow users to set up a profile with name, student ID, batch, section, department, and avatar image. | High     |
| FR-05 | The system shall maintain role-based sessions: Super Admin, Club Executive, and Regular Student.                      | High     |
| FR-06 | The system shall allow the Super Admin to assign or revoke Executive roles for any user.                              | High     |
| FR-07 | The system shall allow users to log out and clear their active session.                                               | Medium   |
| FR-08 | The system shall allow users to edit their profile information after registration.                                    | Medium   |

### 3.1.2 Club Feed & Posts



| ID    | Requirement Description                                                                             | Priority |
|-------|-----------------------------------------------------------------------------------------------------|----------|
| FR-09 | The system shall display a personalized home feed showing posts only from clubs the user follows.   | High     |
| FR-10 | The system shall display a department-wide feed showing posts from all six clubs.                   | High     |
| FR-11 | The system shall allow Club Executives to create posts containing text and/or an uploaded image.    | High     |
| FR-12 | The system shall allow Club Executives to pin up to one post per club at the top of the club feed.  | Medium   |
| FR-13 | The system shall allow all users to react to posts using Like, Fire, and Clap reactions.            | Medium   |
| FR-14 | The system shall allow all users to add comments to any post.                                       | Medium   |
| FR-15 | The system shall allow Club Executives to edit or delete their own posts.                           | Medium   |
| FR-16 | The system shall display posts in reverse chronological order, with pinned posts always at the top. | High     |

### 3.1.3 Club Management

| ID    | Requirement Description                                                                                    | Priority |
|-------|------------------------------------------------------------------------------------------------------------|----------|
| FR-17 | The system shall display a listing of all six CSE department clubs with their name, logo, and description. | High     |
| FR-18 | The system shall allow users to follow or unfollow any club with a single tap.                             | High     |
| FR-19 | The system shall display each club's dedicated page including cover image, bio, executive list, and posts. | High     |
| FR-20 | The system shall allow Club Executives to update their club's bio, cover image, and logo.                  | Medium   |
| FR-21 | The system shall display the list of executive members for each club on the club profile page.             | Low      |

#### 3.1.4 Events Module

| ID    | Requirement Description                                                                                                          | Priority |
|-------|----------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-22 | The system shall allow Club Executives to create events with a title, description, date, time, venue, and optional poster image. | High     |
| FR-23 | The system shall display all upcoming events across all clubs in a dedicated Events screen.                                      | High     |
| FR-24 | The system shall allow users to RSVP to events with Going or Interested status.                                                  | High     |
| FR-25 | The system shall send an automatic push notification to all RSVP'd users 24 hours before an event.                               | High     |
| FR-26 | The system shall allow Club Executives to edit or cancel their own events.                                                       | Medium   |
| FR-27 | The system shall display each event's RSVP count visible to all users.                                                           | Low      |

### 3.1.5 Notifications

| ID    | Requirement Description                                                                                     | Priority |
|-------|-------------------------------------------------------------------------------------------------------------|----------|
| FR-28 | The system shall send an instant push notification to all followers of a club when a new post is published. | High     |
| FR-29 | The system shall send an instant push notification to all followers when a new event is created.            | High     |
| FR-30 | The system shall send a push notification to a user when someone comments on their post.                    | Medium   |
| FR-31 | The system shall maintain a notification center screen showing all past alerts with read/unread status.     | Medium   |
| FR-32 | The system shall allow users to mark all notifications as read.                                             | Low      |

### 3.1.6 Search & Discovery

| ID    | Requirement Description                                                                                    | Priority |
|-------|------------------------------------------------------------------------------------------------------------|----------|
| FR-33 | The system shall provide a search screen where users can search for posts, events, clubs, and other users. | High     |
| FR-34 | The system shall allow filtering of the home feed by department or specific club.                          | Medium   |
| FR-35 | The system shall display search results in real time as the user types.                                    | Medium   |

### 3.1.7 Admin Panel

| ID    | Requirement Description                                                                                     | Priority |
|-------|-------------------------------------------------------------------------------------------------------------|----------|
| FR-36 | The system shall provide the Super Admin with a dedicated dashboard accessible only to admin accounts.      | High     |
| FR-37 | The system shall allow the Super Admin to promote any registered user to a Club Executive role.             | High     |
| FR-38 | The system shall allow the Super Admin to revoke Executive privileges from any user.                        | High     |
| FR-39 | The system shall allow the Super Admin to delete any post or event from any club.                           | High     |
| FR-40 | The system shall display platform-wide statistics to the Super Admin (total users, posts per club, events). | Low      |

### 3.2 Non-Functional Requirements

| ID     | Requirement Description                                                                                                | Type        |
|--------|------------------------------------------------------------------------------------------------------------------------|-------------|
| NFR-01 | The app shall load the home feed within 3 seconds on a standard 4G connection.                                         | Performance |
| NFR-02 | Real-time feed updates shall be reflected on all connected devices within 2 seconds of a post being published.         | Performance |
| NFR-03 | Push notifications shall be delivered within 5 seconds of the triggering event.                                        | Performance |
| NFR-04 | All user passwords must be stored using Firebase Authentication's secure hashing — plaintext storage is not permitted. | Security    |
| NFR-05 | Only authenticated users with verified Executive roles shall be able to create, edit, or delete posts and events.      | Security    |
| NFR-06 | Firebase Security Rules must restrict data read/write access based on user role.                                       | Security    |
| NFR-07 | The app UI shall be fully responsive and usable on screen sizes from 5 to 7 inches.                                    | Usability   |
| NFR-08 | The app shall support both light and dark mode display themes.                                                         | Usability   |
| NFR-09 | All primary user actions (login, post, follow, RSVP) shall be reachable within 3 taps from the home screen.            | Usability   |
| NFR-10 | The APK file size shall not exceed 50 MB to comply with Google Play Store requirements.                                | Portability |
|        |                                                                                                                        |             |

|        |                                                                                                                    |                 |
|--------|--------------------------------------------------------------------------------------------------------------------|-----------------|
| NFR-11 | The app shall be compatible with Android 10 (API 29) and above.                                                    | Portability     |
| NFR-12 | The app shall be maintainable with a modular folder structure separating screens, models, services, and providers. | Maintainability |
| NFR-13 | The application shall achieve at least 95% uptime, dependent on Firebase infrastructure availability.              | Reliability     |

### 3.3 External Interface Requirements

#### 3.3.1 User Interface

- The app shall use a bottom navigation bar with five tabs: Home Feed, Clubs, Events, Notifications, and Profile.
- All screens shall follow Material Design 3 guidelines for consistency and familiarity.
- The app shall display skeleton loading placeholders while data is being fetched from Firestore.
- Error states (no internet, empty feed) shall display friendly messages with actionable prompts.

#### 3.3.2 Hardware Interfaces

- The app shall access the device camera and photo gallery for uploading profile avatars and post images.
- The app shall use the device's push notification system (Android Notification Channels / APNs for iOS).

#### 3.3.3 Software Interfaces

| Interface                | Service         | Purpose                                                             |
|--------------------------|-----------------|---------------------------------------------------------------------|
| Firebase Authentication  | Firebase SDK    | User login, registration, and session management                    |
| Cloud Firestore          | Firebase SDK    | Real-time data storage and synchronization for all app content      |
| Firebase Storage         | Firebase SDK    | Uploading and retrieving images for posts, avatars, and club covers |
| Firebase Cloud Messaging | FCM SDK         | Delivering push notifications to user devices                       |
| Riverpod                 | Flutter package | Reactive state management across the entire application             |

## 4. System Models

### 4.1 Use Case Diagram

*Use Case Diagram showing all three actors and their interactions with the system.*

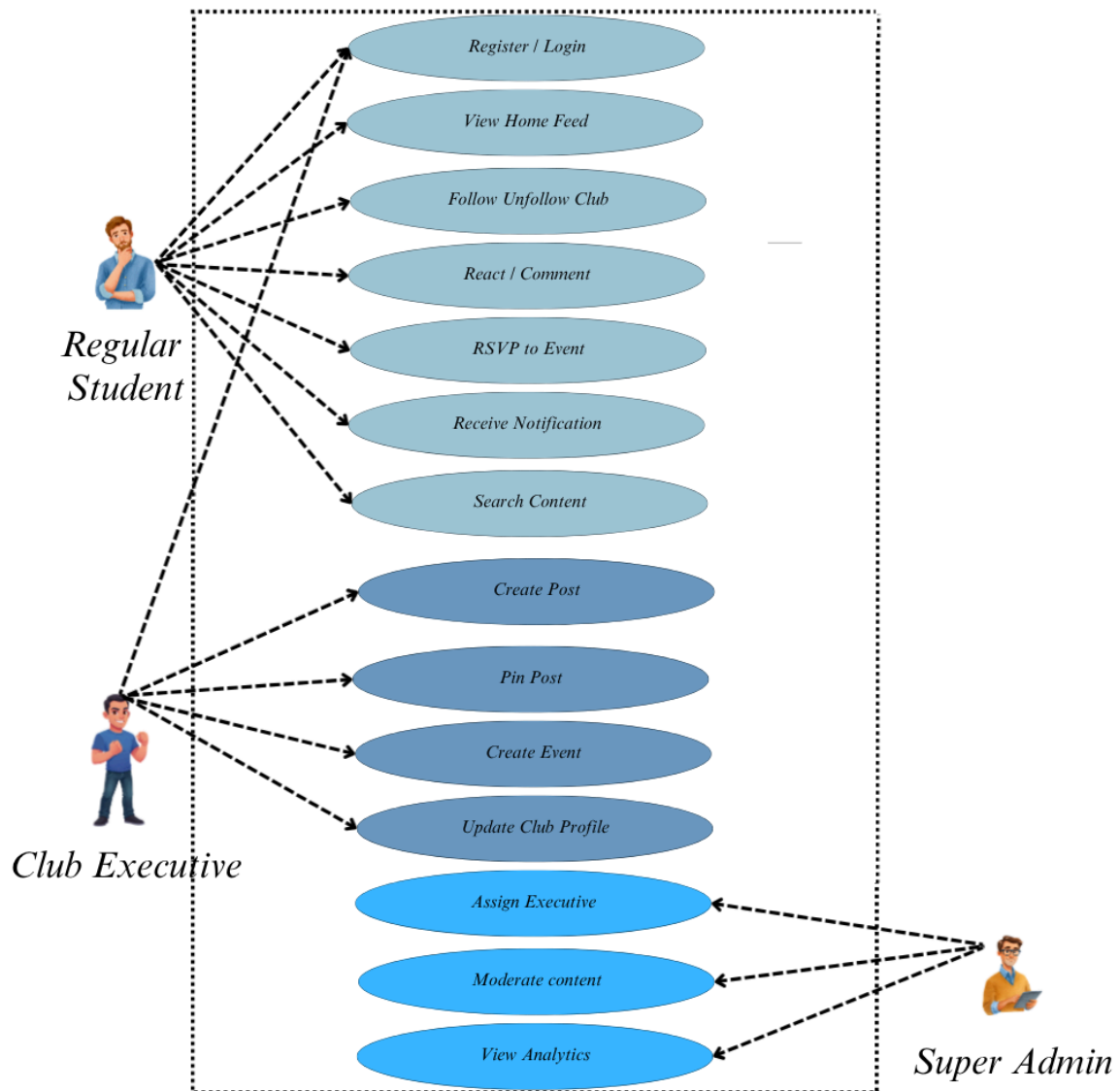


Fig-4.1: Use Case Diagram

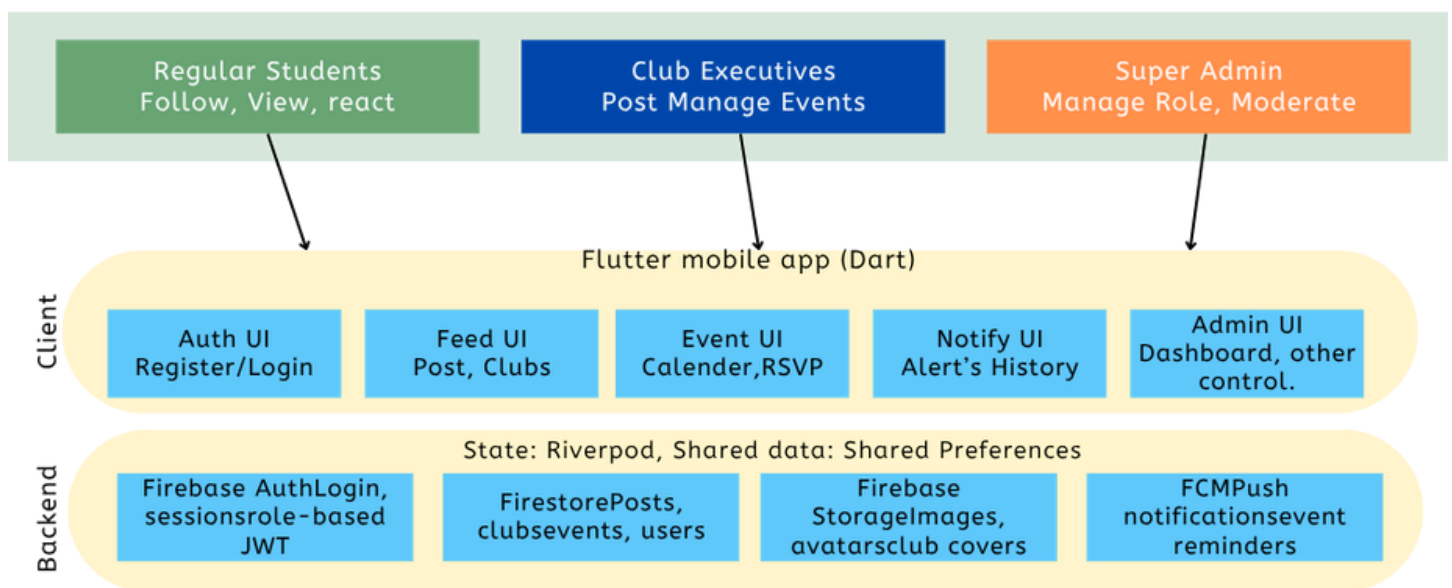
### 4.2 Use Case Summary



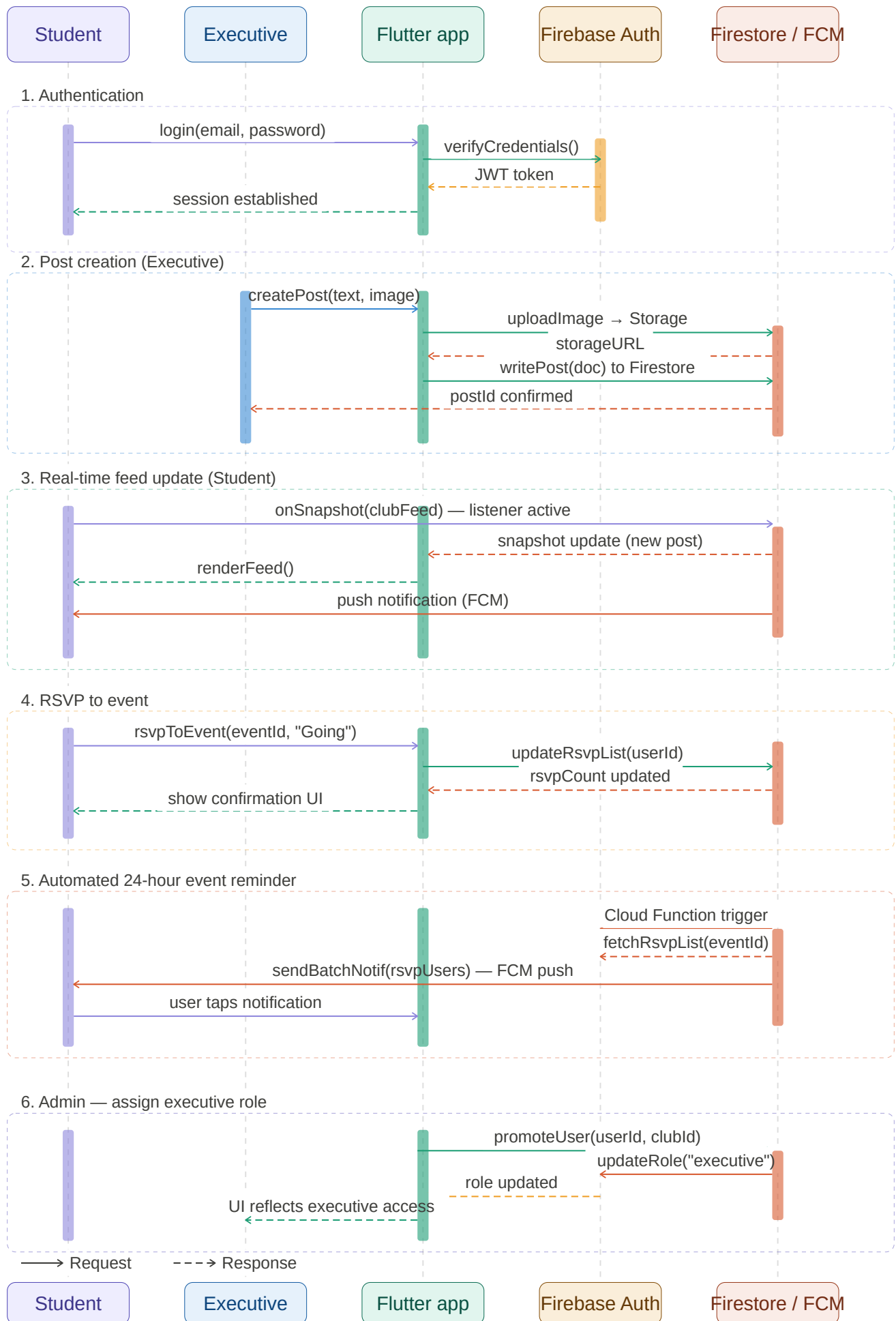
| Use Case ID | Use Case Name        | Primary Actor   | Description                                                                |
|-------------|----------------------|-----------------|----------------------------------------------------------------------------|
| UC-01       | Register Account     | Regular Student | User signs up with university email, password, and profile details.        |
| UC-02       | Login                | All Users       | User authenticates using email and password to access the app.             |
| UC-03       | Follow Club          | Regular Student | User follows a club to receive its posts in their personalized feed.       |
| UC-04       | View Home Feed       | Regular Student | User views a real-time feed of posts from all followed clubs.              |
| UC-05       | Create Post          | Club Executive  | Executive writes a notice and optionally attaches an image for their club. |
| UC-06       | Pin Post             | Club Executive  | Executive pins a critical notice to the top of the club feed.              |
| UC-07       | React to Post        | Regular Student | User reacts to a post using Like, Fire, or Clap.                           |
| UC-08       | Comment on Post      | All Users       | User writes a comment on any post visible to them.                         |
| UC-09       | Create Event         | Club Executive  | Executive creates an event with date, time, venue, and poster.             |
| UC-10       | RSVP to Event        | Regular Student | User marks themselves as Going or Interested for an event.                 |
| UC-11       | Receive Notification | All Users       | User receives push alert for new posts, events, or                         |

|       |                     |                |                                                             |
|-------|---------------------|----------------|-------------------------------------------------------------|
|       |                     |                | comment replies                                             |
| UC-12 | Search Content      | All Users      | User searches for posts, events, clubs, or other users.     |
| UC-13 | Assign Executive    | Super Admin    | Admin promotes a registered student to Club Executive role. |
| UC-14 | Delete Any Content  | Super Admin    | Admin removes any post or event from any club.              |
| UC-15 | Update Club Profile | Club Executive | Executive updates the club's bio, cover image, or logo.     |

#### 4.3 System Architecture Diagram



#### 4.4 sequence Diagram



## 5. Project Management

5.1 Development Tools

| Tool / Technology        | Version    | Purpose                                                   |
|--------------------------|------------|-----------------------------------------------------------|
| Flutter                  | 3.x (Dart) | Cross-platform mobile UI framework                        |
| Firebase Firestore       | Latest SDK | Real-time NoSQL cloud database                            |
| Firebase Authentication  | Latest SDK | Secure user login and session management                  |
| Firebase Storage         | Latest SDK | Cloud storage for images and media files                  |
| Firebase Cloud Messaging | Latest SDK | Push notification delivery                                |
| Riverpod                 | 2.x        | Reactive state management                                 |
| Android Studio           | Latest     | Primary IDE for Flutter development and Android emulation |
| VS Code                  | Latest     | Lightweight code editor with Flutter extension            |
| Git & GitHub             | Latest     | Version control and source code repository                |
| Figma                    | Web        | UI/UX wireframe and screen design                         |
| Postman                  | Latest     | API and Firebase rule testing                             |

5.2 Development Timeline

| Phase               | Week   |                                                                                                    | Deliverable                                        |
|---------------------|--------|----------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Setup & Auth        | Week 1 | Flutter project scaffold, Firebase configuration, login/register screens, role-based session       | Working auth flow with Firestore user creation     |
| Profiles & Clubs    | Week 2 | User profile setup, club listing screen, follow/unfollow functionality, club profile page          | Club browsing and follow system live               |
| Post System         | Week 3 | Post creation UI (text + image), Cloudinary/Firebase Storage upload, Firestore write, feed display | Executives can publish notices; feed renders posts |
| Feed & Interactions | Week 4 | Personalized home feed, department-wide feed, reactions, comment threads, pin feature              | Full post interaction system working               |
| Events Module       | Week 5 | Event creation screen, events listing, RSVP system, event detail page                              | Complete events module with RSVP tracking          |
| Notifications       | Week 6 | FCM integration, push notification triggers, notification center screen, 24-hr event reminders     | Real-time push notifications live                  |
| Admin & Search      | Week 7 | Super Admin dashboard, executive assignment, content moderation, search screen                     | Admin panel and search functional                  |
| Polish & Deploy     | Week 8 | UI refinement, dark mode, edge case handling, bug fixes, Play Store APK preparation                | Production-ready APK ready for submission          |

## 6. Appendix

6.1 Club Reference

| Index | Name                         | Focus Area                                      | Executive Roles                                                        |
|-------|------------------------------|-------------------------------------------------|------------------------------------------------------------------------|
| 1     | Machine Learning Club        | AI, Data Science, Deep Learning                 | President, VP, Secretary, Joint Secretary, Treasurer, Executive Member |
| 2     | Competitive Programming Club | Algorithms, Contests, Problem Solving           | President, VP, Secretary, Joint Secretary, Treasurer, Executive Member |
| 3     | IoT & Robotics Club          | Hardware, Embedded Systems, Automation          | President, VP, Secretary, Joint Secretary, Treasurer, Executive Member |
| 4     | Web Development Club         | Frontend, Backend, Full Stack Web               | President, VP, Secretary, Joint Secretary, Treasurer, Executive Member |
| 5     | Software Development Club    | App Dev, System Design, Software Engineering    | President, VP, Secretary, Joint Secretary, Treasurer, Executive Member |
| 6     | Cyber Security Club          | Network Security, Ethical Hacking, Cryptography | President, VP, Secretary, Joint Secretary, Treasurer, Executive Member |

6.2 Requirement Priority Legend

| Priority | Meaning                                                                                     |
|----------|---------------------------------------------------------------------------------------------|
| High     | Must be implemented for the core system to function.<br>Non-negotiable for v1 release.      |
| Medium   | Important for user experience and completeness.<br>Should be included in v1 if time allows. |
| Low      | Nice to have. Can be deferred to a future version<br>without affecting core functionality.  |

## 6.3 References

- Flutter Official Documentation - <https://flutter.dev/docs>
- Firebase Documentation - <https://firebase.google.com/docs>
- Firebase Cloud Messaging Guide - <https://firebase.google.com/docs/cloud-messaging>
- Riverpod State Management - <https://riverpod.dev>
- Material Design 3 Guidelines - <https://m3.material.io>
- Google Play Store Developer Policy - <https://play.google.com/about/developer-content-policy>
- IEEE Std 830-1998 - IEEE Recommended Practice for Software Requirements Specifications
- Other LLM